

SCHOLAR: Inspectable Local-First Question Answering for Scientific Papers

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Abstract

Researchers often need to verify a paper claim against the exact paragraph, table, or figure that supports it. SCHOLAR is a local-first scientific-document assistant built around that interaction: a user asks a question, receives a cited answer, and clicks a citation to inspect the linked source region in the original PDF. The system preserves page identity and layout coordinates during ingestion, combines lexical and dense retrieval, and records the evidence and verifier actions behind each response. In a retrospective benchmark of 100 questions across 25 papers, fused retrieval reaches gold-page Hit@1 of 0.85; a separate 150-case language-model-based audit reports grounding and citation categories, but is not human ground truth. The demonstration shows how researchers, students, and reviewers can use this inspectable workflow locally, with the final submission providing an installable package and screencast link.

1 Introduction

Researchers, students, and peer reviewers use document assistants to extract experimental results, definitions, and visual evidence from scientific PDFs. These documents are unusually difficult for question answering because a claim may depend on text, a table, a figure, or the spatial relationship between them. Existing retrieval-augmented generation (RAG) systems (Lewis et al., 2020) can produce plausible answers, but following one citation back to the exact supporting region is still awkward, especially when an unpublished paper should remain on the reader’s machine.

We demonstrate SCHOLAR as a concrete interaction: upload a paper, ask a question, receive a cited answer, click a citation, and inspect the corresponding paragraph, figure, or table region in the original PDF. The system keeps the evidence object—not merely a text snippet—attached to the

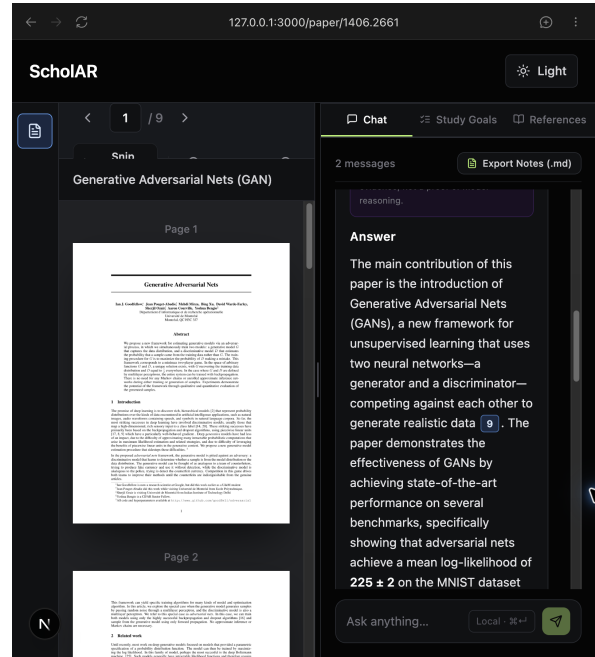


Figure 1: **The interaction the demo exposes.** The workspace places the original PDF beside a cited answer; citation chips are the entry point to page and region inspection. The screencast shows the synchronized jump and evidence-inspection path.

answer and records retrieval, citation, verification, and final-output decisions in an audit trace.

Scientific-document QA benchmarks such as QASPER, SciFact, and M3SciQA show the need to reason over scholarly evidence (Dasigi et al., 2021; Wadden et al., 2020; Li et al., 2024); citation-grounded generation studies evaluate whether retrieved context is used (Gao et al., 2023; Asai et al., 2024); and multimodal document systems show why page layout and visual evidence matter (Banerjee et al., 2024; Faysse et al., 2025). SCHOLAR’s demo contribution is to integrate these concerns into a local interactive workflow in which citations retain spatial identity and can be inspected in the source document.

In this demonstration, we make three contribu-

tions:

1. **Inspectable scientific QA:** A local-first assistant whose answer citations resolve to text bounding boxes, figures, and tables in the original PDF.
2. **Auditable grounding pipeline:** Layout-aware document representation, hybrid retrieval, and deterministic lexical/numerical checks preserve the connection between generated claims and retrieved evidence.
3. **Open interactive system:** A reproducible implementation characterized with retrieval, grounding, and consumer-workstation diagnostics, with the package and screencast supplied as submission artifacts.

The software is MIT-licensed; access details are given in Section 5.

2 System Architecture

SCHOLAR is a local interactive system: a Python/FastAPI backend prepares source-scoped evidence and a Next.js/React frontend presents the answer beside the document. Figure 2 summarizes the three stages that matter to a reader: document representation, evidence retrieval and local answering, and verification and inspection.

2.1 Document representation

When a user provides a scientific PDF, the ingestion engine extracts both structural semantics and visual coordinates:

- A layout-aware parser recovers document hierarchy while isolating figures and tables, with a PyMuPDF fallback.
- Each text or visual element receives a source-scoped identity containing its document, page, and normalized bounding box.
- Figures, tables, and page regions remain available as local crops; ingestion commits atomically so a partial paper is not queryable.

2.2 Evidence retrieval and local answering

BM25 and dense text retrieval are fused with reciprocal-rank fusion; visual crops and full pages provide candidates when a question refers to a figure or table. The application—not the model—owns page numbers and bounding boxes. Retrieved evidence is placed in a structured prompt for a local

Ollama model such as `qwen3.5:9b`; the model emits citation markers referring to evidence items, while the application maps those markers back to source identities. Strict-local mode blocks non-loopback network access during analysis, separating local answering from any explicitly enabled acquisition step.

2.3 Verification and inspection

After generation, SCHOLAR runs a post-generation inspection stage via `verifier_service`:

- lexical overlap and numerical checks flag mismatches between cited claims and supplied evidence;
- bounded remapping or pruning records changes without generating replacement facts; and
- the query, retrieved objects, raw completion, verifier actions, final citations, and timings are stored in a versioned trace that the Evidence Graph view exposes.

The verifier is a routing and inspection aid, not semantic entailment or human-verified ground truth. The user’s central action remains direct inspection of the original PDF region.

3 Interactive Demonstration

The core user experience of SCHOLAR is direct verification: readers can check the evidence behind each citation in the same workspace as the answer. Figure 1 shows the workspace; the screencast supplies the full click-to-region sequence.

3.1 Representative scenarios

The demonstration uses three short scenarios.

Text and numerical verification. The user uploads a paper and asks, “What improvement does the proposed method report over baseline X?” SCHOLAR retrieves the relevant passages and tables, produces a cited answer, and lets the user click the citation to inspect the source page and region.

Visual evidence inspection. The user asks, “What does Figure 3 show about the ablation?” SCHOLAR supplies the figure or table crop with its caption and keeps the citation linked to the original page, allowing the reader to compare the generated description with the source pixels.

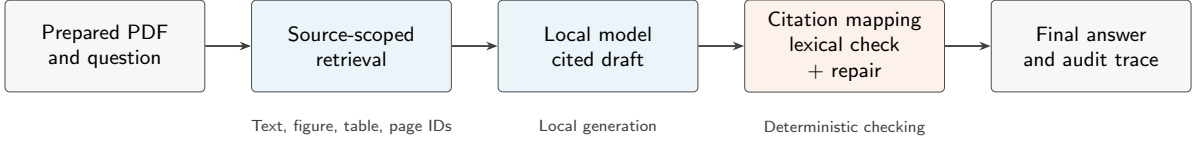


Figure 2: **Three-stage ScholAR pipeline.** A PDF becomes layout-aware evidence objects; hybrid retrieval supplies a local model with source-scoped context; verification and the interface keep the final answer connected to inspectable regions and an audit trace.

Insufficient evidence and audit. The user asks for a fact that is not present in the retrieved paper. The system can expose weak or missing support rather than presenting provenance as proof; the user then opens the trace or Evidence Graph to inspect retrieved candidates, citation origins, verifier actions, and terminal status.

The live demonstration follows the complete path: upload and index a PDF, ask one text or visual question, inspect the cited region, open the graph, and export the trace. Clicking a citation synchronizes the answer and viewer; the interface distinguishes text evidence from figure/table crops and displays the current verification status.

3.2 Intended audience and interface

SCHOLAR targets academic researchers comparing results, peer reviewers checking claims, and graduate students reading technical papers. The value of the demo is not a claim that every answer is correct; it is that the user can move from generated prose to the original evidence with a visible, inspectable path.

Clicking a citation marker triggers an immediate synchronization event:

- the document viewer navigates to the cited page and evidence region;
- the selected text, figure, or table remains distinguishable from the generated answer; and
- the trace and Evidence Graph expose retrieved candidates, citation origins, verifier actions, and final status.

4 Evaluation

4.1 Can the system find relevant evidence?

We evaluate page-level evidence retrieval on a retrospective benchmark of 100 questions across 25 scientific papers. Table 1 compares lexical, dense, and fused retrieval configurations. BM25 places a gold evidence page at rank 1 for 84% of queries

Retrieval Configuration	Hit@1 ↑	Hit@5 ↑	MRR ↑	nDCG@5 ↑
Keyword overlap	0.71	0.91	0.796	0.757
BM25 lexical	0.84	0.97	0.892	0.855
Dense embedding	0.76	0.95	0.838	0.796
BM25 + dense RRF	0.85	0.98	0.905	0.853
BM25 + dense + modality RRF	0.59	0.95	0.734	0.714

Table 1: **Retrospective Page-Retrieval Ablation** on 100 questions across 25 scientific papers. Fusing BM25 and dense retrieval with Reciprocal Rank Fusion (RRF) yields Hit@1 of 0.85 (paired bootstrap difference +0.01, 95% CI $[-0.054, +0.078]$ over BM25). Adding an uncalibrated modality heuristic degrades Hit@1 to 0.59 (-0.26 , 95% CI $[-0.388, -0.136]$).

(MRR 0.892). BM25+dense reciprocal-rank fusion reaches Hit@1 of 0.85 and MRR of 0.905, but its paired bootstrap Hit@1 difference over BM25 is only +0.01 (95% CI $[-0.054, +0.078]$), which includes zero. An uncalibrated modality boost instead drops Hit@1 from 0.85 to 0.59 (-0.26 , 95% CI $[-0.388, -0.136]$). We therefore keep visual evidence available for inspection without treating visual intent as proof that a candidate is relevant.

4.2 Do answers and citations remain grounded?

The completed 150-case LLM-based evaluation provides a categorical diagnostic of answer correctness, evidence grounding, and citation quality. The retained rows identify the evaluator generically, so we do not claim a named judge model; visual labels are excluded because the later visual update introduced a value outside the frozen rubric. On the 113 reference-eligible cases, 59 are correct, 33 partly correct, 20 incorrect, and one not judgeable. Across all 150 cases, grounding is supported in 112 and citation quality is accurate in 110. These are model judgments, not human ground truth.

The deterministic repair ablation remains informative but not a headline quality result: it raises reference-unit overlap F1 from 0.159 to 0.224 while lowering cited-page alignment from 0.670 to 0.618 on the 20 gold-page cases. We therefore treat repair as a diagnostic and preserve the prin-

Dimension	Supported/Correct	Partial	Incorrect/Unsupp.	N/A
Correctness (eligible $n = 113$)	59 (52.2%)	33 (29.2%)	20 (17.7%)	1 (0.9% unjudgeable)
Completeness ($n = 150$)	83 (55.3%)	24 (16.0%)	41 (27.3%)	2 (not judgeable)
Grounding ($n = 150$)	112 (74.7%)	26 (17.3%)	10 (6.7%)	2 (not judgeable)
Citation Quality ($n = 150$)	110 (73.3%)	28 (18.7%)	10 (6.7%)	2 (missing)

Table 2: **LLM-based evaluation results on 150 complement cases.** Correctness is evaluated strictly against the 113 verified eligible references (*37 cases had ineligible reference answers; 1 eligible case was unjudgeable). Grounding and citation quality evaluate local excerpt attribution. The table reports only categories in the frozen rubric; a later visual-label update introduced an out-of-schema category and is excluded. These are model judgments, not human ratings.

inciple that spatially correct evidence matters more than lexical overlap alone; the paired table is in Appendix C.

4.3 Can the complete path run locally?

Across 100 valid requests on an Apple M3 Pro with 18 GB unified memory and local Ollama, end-to-end latency has $p_{50} = 27.34$ s, $p_{90} = 41.61$ s, and $p_{95} = 149.54$ s. Retrieval has median 523 ms; generation has median 26.59 s and $p_{95} = 37.48$ s; verification and trace writing have median 15 ms and $p_{95} = 24$ ms. These observations establish feasibility on the named workstation, not interactive responsiveness at scale.

Taken together, these results support a narrow claim: the current prototype retrieves useful evidence and runs locally on the tested workstation. They do not establish generalization or human benefit.

5 Availability and Reproducibility

Access. SCHOLAR is MIT-licensed; its source is at <https://github.com/prithvi-kaizen/Scholar>. The final submission will add a signed-out-accessible, versioned ZIP URL with a SHA-256 checksum and a screencast URL (no longer than 2.5 minutes). The release contains a self-authored example PDF; private papers, caches, traces, and model weights are excluded.

Tested environment. The verified profile uses Python 3.12, Node.js 20/npm 10, Ollama, and a Q4_K_M qwen3.5:9b model on an Apple M3 Pro with 18 GB unified memory.

6 Conclusion

We presented SCHOLAR, a local-first scientific-paper assistant built around one interaction: ask

a question, then inspect the exact source region attached to each citation. Layout-aware evidence objects, hybrid retrieval, deterministic checks, and an execution trace support that interaction without turning provenance into a claim of factual entailment. The retrospective diagnostics show useful retrieval and feasible local operation, while also showing why lexical repair and model judgments require caution. The demo’s contribution is an inspectable workflow for researchers, reviewers, and students; its package and screencast must make that workflow directly testable.

Limitations

The current system and evidence have four clear limits.

Layout and parsing. The dual-engine ingestion pipeline works best on modern two-column preprints. Degraded scans, multi-page tables, intricate vector graphics, and formulas that depend on custom fonts can produce misaligned boxes or incomplete text and table extraction.

Verifier scope. The verifier checks lexical overlap, n-gram containment, and numerical patterns. It does not establish semantic entailment or prove a mathematical statement. It can therefore miss a conceptual error that reuses source vocabulary, or remove a valid paraphrase whose wording differs substantially from the source.

Hardware and latency. The measured profile covers one Apple M3 Pro workstation with 18 GB unified memory. End-to-end latency is $p_{50} = 27.34$ s and $p_{95} = 149.54$ s under uncontrolled cache conditions. Other devices may be slower or may not fit the local model; we make no claim about responsiveness at scale.

Evidence scope. The retrieval benchmark contains 100 retrospective questions over 25 papers, so its results are not population estimates. The 150-case language-model audit is not human ground truth, and no real researcher workflow or user benefit has yet been measured.

Ethical Considerations

Privacy and local execution. SCHOLAR is intended for unpublished manuscripts, review drafts, and proprietary notes. In strict-local analysis mode, the network policy blocks non-loopback requests. Users must still distinguish that analysis boundary from any explicitly enabled document, model, or reference acquisition.

Copyright and redistribution. The demonstration package and public repositories do not redistribute copyrighted publications without authorization. Benchmarks and tutorial examples use open-access preprints distributed under Creative Commons or arXiv non-exclusive licenses.

Evaluators and consent. The five-rater offline study protocol was prepared, but no ratings had been returned at the manuscript freeze. If recruitment proceeds, evaluators will provide informed consent and use pseudonymous identifiers (e.g., R01–R05); public aggregates will not expose contact details or identifiable comments.

External models. Secondary evaluation with an external model requires explicit operator authorization and sets `store=false`. Users are told that external APIs are not zero-retention environments.

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A Prompt Templates and Generation Protocol

The answer generation pipeline utilizes a structured system prompt designed to enforce factual citations and prevent hallucinated references.

You are a precise scientific research assistant. You are given a user question and a numbered set of verified evidence excerpts from a scientific paper.

Answer the question strictly using the provided evidence excerpts. Every factual claim, quantitative value, or mathematical statement MUST be followed immediately by one or more citation markers referencing the exact excerpt number, e.g., [1] or [1, 2].

Do not introduce external knowledge or facts not present in the excerpts. If the provided excerpts do not contain sufficient evidence to answer the question, state clearly: "The provided paper sections do not contain sufficient information to answer this question."

Evidence items presented in the prompt follow a strict template:

[Excerpt 1] (Page 4, Section 3.2)
"We train our model using Adam with $\text{betal}=0.9$, $\text{beta2}=0.999$, and a learning rate of $2e-4$..."

B Artifact Schema and Execution Trace

Every question answering interaction produces a validated JSON execution trace recorded under `backend/data/traces/`. The trace structure includes:

- `query_id`: Unique UUID for the transaction;
- `paper_id`: Ingested document identifier;
- `raw_query`: Raw natural language question;
- `retrieved_chunks`: Ranked list of chunk IDs, fusion scores, page numbers, and normalized bounding boxes $[x_0, y_0, x_1, y_1]$;
- `raw_completion`: Unmodified text emitted by the generator;
- `verifier_actions`: List of remapping or pruning operations executed by the consistency verifier;

- `final_answer`: Verified answer with synchronized citation chips;
- `latency_breakdown`: Latency in milliseconds across retrieval, inference, and verification.

C Evaluation Protocols and Sampling Breakdowns

This section records the sampling strata, case-eligibility partitions, repair diagnostic, and language-model audit used to characterize the system. The human-study protocol is included for transparency; no human ratings had been returned at the manuscript freeze, so it is not reported as an outcome.

Question Category	Easy	Medium	Hard	Total
Direct text lookup	1	5	4	10
Multi-passage synthesis	4	10	9	23
Figure / visual evidence	2	2	4	8
Mathematical mechanism	1	1	1	3
Table / quantitative	2	2	2	6
Total Stratified Sample	10	20	20	50

Table 3: **Stratified Sampling Breakdown of the 50-Question Human Evaluation.** Five independent raters evaluate the same 50 questions across 10 papers ($n = 250$ total ratings planned).

Automatic Diagnostic	Off	On	Paired Change (95% CI)
Reference-unit overlap F1	0.159	0.224	+0.065 [+0.042, +0.091]
Automatic key-term coverage	0.759	0.742	-0.017 [-0.038, 0.000]
Citation-page alignment ($n = 20$)	0.670	0.618	-0.052 [-0.091, -0.015]
Pipeline SUCCESS status	28/30	30/30	+2/30

Table 4: **Retrospective Diagnostic of Post-Generation Repair** on 30 paired questions across 19 papers with identical raw completions. Repair raises lexical overlap F1 but reduces cited-page alignment on gold-page cases. Overlap measures lexical surface matching, not factual correctness.

D Detailed Hardware Profiling

Profiling was conducted on an Apple M3 Pro (12 CPU cores, 18 GPU cores, 18 GB unified memory, macOS 15.1). The local model backend was Ollama v0.5.12 serving qwen3.5:9b with Q4_K_M quantization. The audited 100-request local profile is reported in Section 4; it establishes feasibility on this workstation, not throughput or responsiveness at scale.